Oscilloscope



Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Cathode Ray Oscilloscope (CRO); CRO; Oscilloscope

What it is used for

An oscilloscope displays electrical signals as waveforms over time, allowing users to measure signal level, timing, frequency, and noise. Correct probe selection, grounding, voltage limits, and measurement setup are essential for reliable results and protection against electric shock or instrument damage.

Main parts

- Display shows the measured waveform and measurement information.
- Input channels connect probes to the instrument.
- Probe attenuates or transfers the signal to the input channel.
- Vertical controls adjust voltage scale and channel position.
- Time-base controls adjust the horizontal time scale.
- Trigger controls stabilise repeating waveforms on the display.
- Probe ground lead provides the reference connection where applicable.
- Measurement controls and menu functions calculate signal parameters.
- Power input and protective earth connection supply power and provide grounding as designed.

Before you start

1. Only trained users should measure electrical circuits with an oscilloscope.
2. Read the instrument and probe manuals and identify the circuit voltage, signal type, and measurement category.
3. Inspect the oscilloscope, probes, leads, connectors, and power cable for damage.
4. Confirm the probe attenuation setting matches the instrument setting and the probe is rated for the expected voltage and frequency.
5. Use a suitable differential or isolated probe when required; never assume a standard ground clip is isolated.
6. Confirm the circuit reference and protective-earth requirements before connecting any probe ground.
7. Keep the bench dry and remove unnecessary conductive objects from the measurement area.

How to use

1. Switch on the oscilloscope and allow it to complete its startup checks.
2. Connect the correct probe to the selected input channel and set the matching attenuation factor.
3. Configure the vertical scale, time base, coupling, and trigger according to the signal being measured.

4. With the circuit de-energised where required, connect the probe tip to the test point and the reference lead only to an approved reference point.
5. Energise the circuit only after checking all connections and confirming the setup is safe.
6. Adjust the trigger and scales until the waveform is stable and clearly visible.
7. Use cursors or automatic measurement functions to read voltage, period, frequency, or other required parameters.
8. Compare results with the expected signal range and note any clipping, noise, distortion, or unexpected transients.
9. De-energise the circuit before moving probe connections when required by the risk assessment.
10. Remove the probe tip and reference lead safely, then switch off the instrument if no longer needed.
11. Save waveform data or record readings and report abnormal results or instrument faults.

⚠ Safety

- Never exceed the voltage, frequency, or measurement-category ratings of the oscilloscope or probe.
- Do not connect a standard earth-referenced probe ground clip to a live point that is not at earth potential.
- Never remove the oscilloscope protective earth to make an unsafe measurement.
- Use correctly rated differential probes or approved isolation methods for floating or high-side measurements.
- Keep hands away from exposed live conductors and use suitable PPE when required.
- Disconnect power and discharge stored energy before changing circuit connections whenever the procedure permits.

Care & storage

- Switch off and disconnect the instrument before cleaning its exterior.
- Clean the display and housing with manufacturer-approved materials; keep liquids away from openings.
- Inspect probes, tips, ground leads, connectors, and cables regularly.
- Store probes without sharp bends and protect the instrument from dust, moisture, and impact.
- Arrange calibration or service if readings are unreliable, controls fail, or the instrument is damaged.

Common mistakes to avoid

- Using the wrong probe attenuation or an unsuitable voltage rating.
- Connecting the probe ground clip to an unsafe reference point.
- Changing probe connections on an energised circuit without an approved procedure.
- Misreading an unstable waveform because trigger and time-base settings are incorrect.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.